MA2108(3)

Test 2

Wed. 07/10/2026

Time: 45 minutes Total marks: 20

1. You are given that $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation and that $T(1,1) = (1,0)$ and $T(1,2) = (0,1)$. Find a formula for $T(x,y)$.

[5 marks]

2. Let

$$A = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 2 & 0 & 1 & -2 \\ 0 & 1 & 0 & 4 \\ 2 & 1 & 1 & 2 \end{pmatrix}.$$

   (a) Find a basis for the column space of A, $C(A)$.
   (b) Find a basis for the row space of A, $R(A)$.

[2 + 2 = 4 marks]

3. Let $V = \mathbb{R}^2$ and $F = \mathbb{R}$. Define a homomorphism $T : V \to V$ as

$$T(x_1, x_2)^T = (x_1 - x_2, x_1 + x_2, x_1)^T.$$

   Verify Sylvester's law.

[6 marks]

4. Find an orthonormal basis of the subspace W of $\mathbb{R}^3$, spanned by

$$v_1 = (1,1,0)^T \quad \text{and} \quad v_2 = (-1,0,1)^T.$$

[5 marks]

[5 + 4 + 6 + 5 = 20 marks]

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